BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Waves transfer:

(A) Velocity
(B) Energy
(C) Wavelength
(D) Frequency

Q2. The unit of intensity of sound is :

(A) Wm^{-2}
(B) Wm^{2}
(C) Wm^{-1}
(D) Wm

Q3. Snell's law is :

(A) n = sin r / sin i
(B) n = sin r
(C) n = sin i / sin r
(D) n = sin i

Q4. The S.I. unit of electric field intensity is :

(A) NC^{-1}
(B) NC^{-1}
(C) NC
(D) Nm

Q5. Capacitance of any capacitor is equal to :

(A) C = Q / V
(B) C = V / Q
(C) C = QV
(D) C = 1 / QV

Q6. Formula of e.m.f is equal to :

(A) E = Q / W

(B) E = Q / t

(C) E = W / t

(D) E = W / Q

Q7. The unit of electric power is :

(A) Ohm

(B) Watt

(C) Farad

(D) Joule

Q8. For an ideal transformer:

(A) Pp < Ps

(B) Pp > Ps

(C) Pp = Ps

(D) Pp ≠ Ps

Q9. The logical operation performed by this gate is : (gate symbol shown in the paper)

(A) AND

(B) NOR

(C) OR

(D) NAND

Q10. Boolean expression of NOR gate is :

(A) X = A.B (with overline over A.B)

(B) X = A+B (with overline over A+B)

(C) X = A+B

(D) X = A.B

Q11. Number of components of computer based information system is :

(A) 3

(B) 4

(C) 5

(D) 6

Q12. Generally an atom is represented by the symbol :

(A) $^{A}/_{z}X$

(B) ^{z}X (partially obscured by watermark)

(C) ^{z}X / (Z+N)

(D) X with ^{A} over (A+Z) (partially obscured by watermark)

Physics (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i Define quality of sound.

ii Define spring constant "K" and write down its unit.

iii Define restoring force.

iv Prove that: $v = f \lambda$

v How does loudness depend on the area cf vibrating body? (printed "cf" in the source scan)

vi Define pole and centre of curvature in spherical mirrors.

vii Differentiate between the focus of convex mirror and concave mirror.

viii What is meant by distance formula?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i Define ohmic and non-ohmic conductors.

ii Define electric power and write its formula.

iii Define electromotive force and write its unit.

iv Define mutual Induction.

v State the right hand rule.

vi Draw the symbolic diagram for NAND gate and write its truth table.

vii Define truth table.

viii What is an electric current and write its formula?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i Define electric field intensity and write its equation.

ii Define unit of capacitance.

iii Write two uses of capacitors.

iv Explain the software component of computer based information system.

v What is optical fibre? Write its one advantage for transmission of data.

vi Write name of two web browsers.

vii Define cosmic radiations and write its source.

viii What do you understand by half life of a radioactive element?

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Prove that motion of a simple pendulum is simple harmonic motion.

Q5.(b). (Marks 5) Two capacitors of capacitances 12μF and 6μF are connected in parallel with a 12V battery. Find the equivalent capacitance of the combination. Find the charge across each capacitor.

Q6.(a). (Marks 4) What is Solenoid? Explain the magnetic field of a Solenoid?

Q6.(b). (Marks 5) An object 10 cm in front of a convex mirror forms an image 5 cm behind the mirror. What is the focal length of the mirror?

Q7.(a). (Marks 4) What is meant by nuclear transmutation? Write the general equation of beta decay and also write an example.

Q7.(b). (Marks 5) The resistance of a conductor wire is 10 MΩ. If the potential difference of 100 volts is applied across its ends, then find the value of current passing through it in mA.

GROUP – II

Physics (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which of the gate is used to convert one logic level into opposite logic level :

(A) AND gate
(B) OR gate
(C) NOR gate
(D) NOT gate

Q2. How will be electric lines of force where intensity of electric field is maximum :

(A) Wider
(B) +ve to -ve
(C) Narrow
(D) -ve to +ve

Q3. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire :

(A) Will be zero
(B) Remains the same
(C) Decreases
(D) Increases

Q4. In which state of matter longitudinal waves move faster:

(A) Liquid
(B) Solid
(C) Gas
(D) Liquid and gas

Q5. Isotopes of hydrogen are :

(A) 2
(B) 3
(C) 4
(D) 5

Q6. Speed of sound in hydrogen at 0°C is :

(A) 1290 ms^{-1} (B) 972 ms^{-1}

(C) 1498 ms^{-1} (D) 317 ms^{-1}

Q7. Which of these is not a web browser:

(A) Youtube (B) Chrome

(C) Mozilla firefox (D) Safari

Q8. The refractive index of medium is equal to :

(A) n = 1/cv (B) n = v/c

(C) n = cv (D) n = c/v

Q9. The image formed by convex mirror is :

(A) Real and erect (B) Inverted and real

(C) Erect and virtual (D) Inverted and virtual

Q10. One micro ampere is equal to :

(A) $10^{-3}A$ (B) $10^{-6}A$

(C) $10^{-9}A$ (D) $10^{-12}A$

Q11. According to Coulomb's law what happens to force of attraction of two oppositely charged objects as their distance of separation increases :

(A) Decreases (B) Increases

(C) Remains unchanged (D) Cannot be determined

Q12. In parallel combination of resistors, each resistor will have the same :

(A) Resistance (B) Voltage and current

(C) Voltage (D) Current

Physics (Lahore, Tenth 2021) (Paper - II) (Essay Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 1.45 hrs | **Maximum Marks:** 48

QPART - I.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

i Prove that: $v = f \lambda$

ii What is a spring constant? Write its unit.

iii Define restoring force.

iv Describe how the sound is produced?

v Calculate the frequency of sound wave of speed 340 ms^{-1} and wave length 0.5 m.

vi What are spherical mirrors? Name their two types.

vii What is mirror formula ? Write down its equation.

viii State the laws of refraction.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

i Define electric power and write its unit.

ii State Joule's law and write its formula.

iii Define electro-motive force and write its unit.

iv State Lenz's law.

v What is meant by mutual induction?

vi Define OR gate and write its truth table.

vii Write the name of logic operations.

viii Draw a symbolic diagram for NAND gate and write its truth table.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

i Write two characteristics of electric field lines.

ii Define potential difference and write its formula.

iii Define capacitance and write its unit.

iv What is internet?

v What is meant by data?

vi Write two advantages of electronic mail.

vii Write the general equation of gamma decay and write an example.

viii What is meant by ionization power?

QPART - II. Note: Attempt any TWO questions.

Q5.(a). (Marks 4) Prove that vibratory motion of a mass attached to a spring is S.H.M.

Q5.(b). (Marks 5) A capacitor holds 0.06 coulombs of charge when fully charged by a 9 volt battery. Calculate capacitance of the capacitor.

Q6.(a). (Marks 4) What is transformer? Explain its construction and types.

Q6.(b). (Marks 5) An object and its image in a concave mirror are of a same height, yet inverted, when the object is 20 cm from the mirror. What is focal length of mirror?

Q7.(a). (Marks 4) Define natural radioactivity. Also write any three characteristics of alpha particles.

Q7.(b). (Marks 5) Calculate one month cost of using 50W energy savers for 8 hours daily in your study room. Assume that the price of a unit is Rs 12.